



### Folding phone by Xiaomi

Xiaomi's president Lin Bin, has shared a video of a squarish tablet, foldable into a phone-shaped form. <https://digit.in/feb19-21>



### AI co-piloted race cars

McLaren concept AI system to help with racing strategy and keep the car's systems at their peak. <https://digit.in/feb19-22>

## HOW WE TESTED

While we have had established test processes at the Digit Labs for all kinds of audio categories for years now, 2018 saw the addition of an entirely new category into the market – the True Wireless IEMs. The sheer number and distinctive features made it impossible to club them together with wireless IEMs. However, this entailed the creation of a new test process with the uniqueness of the category in mind.

Now, whenever we talk about wireless audio, we do hear a few distinctly snobbish scoffs go around the room – after all, how can it dream of matching up to the quality of its wired brethren? However, Bluetooth audio codecs over the years have constantly improved, bringing wireless audio to a level where you have to take it seriously, certainly more than a gimmick. The same is applicable for true wireless IEMs. Hence it was important to pick a source that was capable of handling a codec that was easily available to most of our users, yet held its own in terms of being an unbiased, neutral audio source. We decided to stick to our tried and tested FiiO X5 3rd Gen that ships with aptX audio support.

Now, when testing any audio equipment, as we've explained in our previous comparison tests, we go after neutrality. While almost every single person who listens to music certainly has a preference for a certain genre or type, we've always given preference to the sound being 'true to source', i.e. what it was intended to sound like by those who produced it. For such purposes, it is ideal to use a frequency response curve generator. However, nothing but the very best ones would truly serve the purpose. Instead, a reference device with a flat audio output is often used to be compared against each device. For this test, we used the Audeze LCD-i4 and the Campfire Audio Atlas in combination to judge tonal balance, bass and treble response and a lot of other factors across our collection of test tracks.

In terms of the tracklist, we've seen very few changes. Some of the tracks on this list have been with us for years, and as a result, are familiar like the back of our very hands. Each track is focussed on one crucial aspect of listening to music on IEMs. For instance, tracks with high instrument density are used to judge detail reproduction – is the IEM able to reproduce individual instruments accurately in a track

where a lot of instruments overlap? Similarly, other tracks are used to judge bass sharpness, separation and soundstaging, and a number of other factors. None of the selected tracks was under 96kHz sampling rate of 24bit bit depth.

While build, design and fit are highly important for any kind of device, they hold a particularly important place in the evaluation of true wireless IEMs. Irrespective of how good a particular IEM sounds, if it doesn't fit right, or cannot withstand the elements when it needs to, it's pretty much useless. Additionally, it should be designed to be usable – you don't want to stick something in your ear only to resort to weird acrobatics each time you have to change a track or adjust the volume. We also look at the material that has been used to build the device, with factors such as robustness and fingerprint retention playing a role here. Another thing that is of utmost importance is the battery backup and, consequently, the charging case.

True wireless IEMs inevitably ship with a charging case, which also doubles up as the hard case for the units. These units solve two problems – they make losing your individual left or right units more difficult, and they offer additional battery backup, which is an essential feature since most of the True Wireless IEMs currently in the market suffer from poor battery life. You don't want the latest tech if it spends half its life in a charging pod, do you?

As an added bonus, a number of these IEMs come with extra features. For instance, some of them track your steps or activity to double up as fitness trackers. Some provide noise cancellation or audio-transparency as a feature, either locking in or letting in audio depending on what you require at the moment.

Keep in mind that the overall score, and consequently, the winner, is based on a combination of all of these factors. In real life situations, one person is rarely going to tick the exact same set of boxes that we have, especially in music preference. So if you're planning on picking a set of true wireless IEMs based on this test, our advice would be to look for the particular parameters, such as a song from the genre closest to what you like or durability as a feature if you're looking for a workout IEM, and then make the call. In case you're still stuck, you could always email us at [editor@digit.in](mailto:editor@digit.in) with your queries.

bogged down during the initial setup – feel free to skip those and get right to the music. The sound on the Dash Pro comes across as true and clear across genres. There's a slight bias towards the lows which can sound a bit muddy at times, but for most types of music, the Dash Pro is a rock solid performer. The mids and highs are also handled well. The passive noise cancellation on these IEMs is quite good due to the amazing fit. Overall, the Dash Pro caters to the more popular tastes in music.

**Verdict:** Bragi wanted to put a computer in your ear with the Dash Pro. Thanks to the multitude of features that the Bragi comes with and how efficiently and flawlessly they

work, the earbuds almost feel like one. Even with a few minor hiccups here and there with things like automatic activity detection and song playback, the Dash Pro is still a more interesting product than anything we've tested in Bluetooth audio in a long time. Even without all of that, it still makes a pretty strong case for itself with sheer performance. Is it worth the near 30k price? Depends on how important these additional features are for you.

### SONY WF SP700N

**Build and design:** There are quite a few things that seem to have gone wrong with the SP700N's build. For starters, the moment you take them out of the box, it is evident that put-

ting them back in is going to be a task. You have to tuck the hooks inside first. Even when you do that, you need to make sure that the connectors are in place or the earbuds won't charge. The protruding design makes it difficult to lie down on your side. The only buttons on the Sony are on the side and they're pretty hard to press – definitely not as seamless as the touch interface on the Dash Pro or the E8. The fact that it omits volume controls on the earbuds is a huge miss. At 8 hours overall, the cumulative battery life is also unacceptable.

**Performance:** According to Sony, the bass is the highlight of this device. We would agree, with the highest scores in the performance section for